Model Interpretability

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Questions to discuss

1. What is simple and multiple linear regression?
2. What are the assumptions of linear regression?
3. What are the statistical inferences we can make from a linear regression model?
4. How can we interpret and understand the model predictions using SHAP values?

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What is simple and multiple linear regression?

- Linear regression is a way to identify a relationship between the independent variable(s) and dependent variable
- We can use these relationships to predict values for one variable for the given value(s) of the other variable(s)
- The variable which is used in prediction is termed as independent/explanatory/regressor, and the predicted variable is termed as dependent/target/response variable.
- In the case of linear regression with a single explanatory variable, the linear combination can be expressed as

  response = intercept + constant * explanatory variable

- Multiple linear regression is just the extension of the concept of simple linear regression with one variable
- It can be represented by

  target = intercept + constant1*feature1 + constant2*feature2 + constant3*feature3 + ...

- The model aims to find the constants and intercept such that the hyperplane is the best fit

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What are the assumptions of linear regression?

Assumption	How to test	How to fix
No multicollinearity in independent variables	Heatmaps of correlations or VIF (Variance inflation factor)	Remove correlated variables
There should be a linear relationship between dependent and independent variables	Plot residuals vs. fitted values and check the plot	Transform variables that appear non-linear (log, square root, etc.)
The residuals should be independent of each other	Plot residuals vs. fitted values and check the plot	Transform variables (log, square root, etc.)
Residuals must be normally distributed	Plot residuals or use Q-Q plot	Non-linear transformation of the independent or dependent variable
No heteroscedasticity, i.e., residuals should have constant variance	Use statistical test (like goldfeldquandt test)	Non-linear transformation of the dependent variable or add other important variables

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Testing Multicollinearity using VIF

- Multicollinearity occurs when predictor variables in a regression model are correlated.
- When we have multicollinearity in the linear model, the coefficients that the model suggests are unreliable.
- We can detect or test for multicollinearity using the Variance Inflation Factor or VIF.
- Variance inflation factors measure the inflation in the variances of the regression parameter estimates due to collinearities that exist among the predictors.
    - If VIF is 1, then there is no correlation between the selected predictor and the remaining predictor variables, and hence, the variance of its coefficient is not inflated at all.
- General Rule of thumb:
    - 1 < VIF <= 5: There is low multicollinearity
    - 5 < VIF <= 10: There is moderate multicollinearity
    - VIF > 10: There is high multicollinearity

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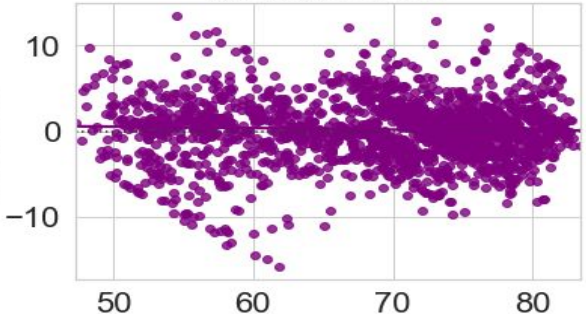
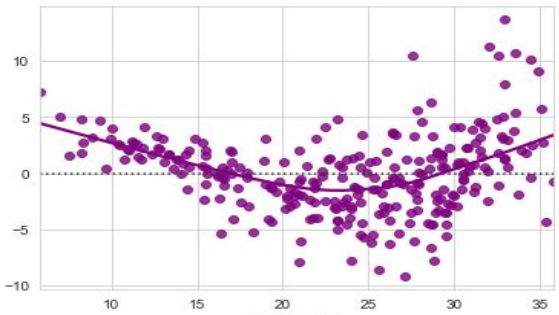
Testing Linearity and Independence using residuals vs fitted values plot

- Predictor variables must have a linear relation with the dependent variable.
- If the residuals are not independent, then the confidence intervals of the coefficient estimates will be narrower and make us incorrectly conclude a parameter to be statistically significant.
- We can check for linearity and independence by checking a plot of fitted values vs residuals.
    - If they don't follow any pattern, then we say the model is linear and residuals are independent.
    - Otherwise, the model is showing signs of non-linearity and residuals are not independent.



No pattern spotted



Some non-linearity spotted

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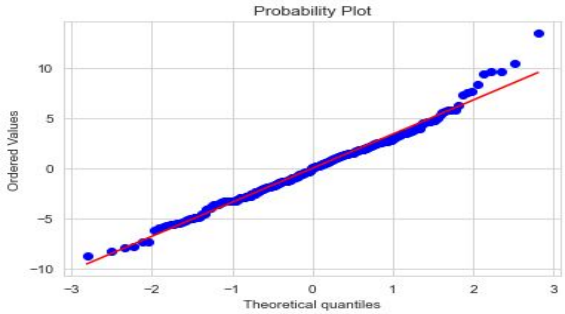
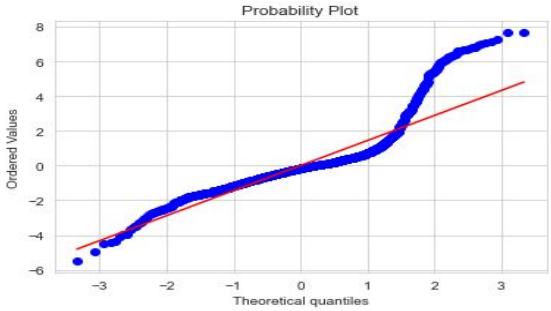
Testing Normality using QQ plots

- If the error terms are not normally distributed, the confidence intervals of the coefficient estimates may become too wide or narrow.
- Non-normality suggests that there are a few unusual data points that must be studied closely to make a better model.
- The shape of the histogram of residuals can give an initial idea about the normality.
- We can also check normality via a Q-Q plot of residuals.
  - If the residuals follow a normal distribution, they will make a straight line plot, otherwise not.



Close to normal



Not normal

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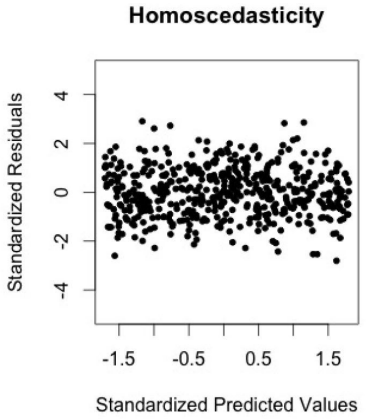
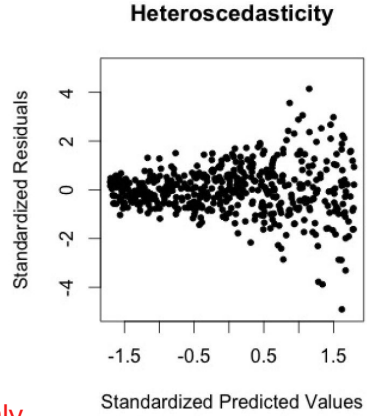
Testing Homoscedasticity using goldfeldquandt test

- If the variance of the residuals is symmetrically distributed across the regression line, then the data is said to be homoscedastic. Else, they are heteroscedastic.
- Generally, non-constant variance arises in presence of outliers.
- The residual vs fitted values plot can be looked at to check for homoscedasticity.
    - In the case of heteroscedasticity, the residuals can form an arrow shape or any other non-symmetrical shape.
- The goldfeldquandt test can also be used.
    - If we get a p-value > 0.05 we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.
    - Null hypothesis: Residuals are homoscedastic
    - Alternate hypothesis: Residuals have heteroscedasticity





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What are the statistical inferences we can make from a linear regression model?

- **Confidence interval**: Give us the 95% confidence interval for the coefficient
  - The 95% confidence interval for mediaType_TV is [-0.625, -0.494].
  - This means we are 95% confident that the true effect of the mediaType_TV on the rating lies between -0.625 and -0.494, holding all other variables constant.
- **p-values**: Give us an idea of whether a particular predictor variable has a statistically significant effect on the target variable or not. If p-value > 0.05, the variable is not statistically significant
  - The p-value for mediaType_TV is 0.0, indicating that it is statistically significant in predicting rating.
- Similarly, we can draw inferences for all other variables.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 rating   R-squared:                       0.718
Model:                            OLS   Adj. R-squared:                  0.717
Method:                 Least Squares   F-statistic:                     482.7
Date:                Thu, 17 Jul 2025   Prob (F-statistic):               0.00
Time:                        12:11:37   Log-Likelihood:                -2340.4
No. Observations:                4566   AIC:                             4731.
Df Residuals:                    4541   BIC:                             4891.
Df Model:                          24
Covariance Type:            nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                          2.7875      0.033     84.367      0.000       2.723       2.852
eps                            0.0201      0.001     18.671      0.000       0.018       0.022
duration                       0.0123      0.000     26.100      0.000       0.011       0.013
watched                        0.0002   6.16e-06     31.750      0.000       0.000       0.000
watching                       0.0031      0.000     19.918      0.000       0.003       0.003
years_running                 -0.0762      0.009     -8.944      0.000      -0.093      -0.059
mediaType_Movie               -0.3078      0.032     -9.550      0.000      -0.371      -0.245
mediaType_Music Video         -0.2987      0.030     -9.971      0.000      -0.357      -0.240
mediaType_OVA                 -0.3016      0.030    -10.092      0.000      -0.360      -0.243
mediaType_Other               -0.2607      0.035     -7.548      0.000      -0.328      -0.193
mediaType_TV                  -0.5598      0.033    -16.781      0.000      -0.625      -0.494
mediaType_TV Special          -0.1816      0.039     -4.657      0.000      -0.258      -0.105
mediaType_Web                 -0.4164      0.031    -13.523      0.000      -0.477      -0.356
contentWarn_Yes               -0.1786      0.020     -8.745      0.000      -0.219      -0.139
studio_primary_J.C. Staff     -0.2017      0.041     -4.865      0.000      -0.283      -0.120
studio_primary_MADHOUSE       -0.2532      0.043     -5.844      0.000      -0.338      -0.168
studio_primary_OLM            -0.4121      0.052     -7.941      0.000      -0.514      -0.310
studio_primary_Others         -0.2993      0.024    -12.445      0.000      -0.346      -0.252
studio_primary_Studio Deen    -0.1657      0.048     -3.432      0.001      -0.260      -0.071
studio_primary_Studio Pierrot -0.2740      0.049     -5.591      0.000      -0.370      -0.178
studio_primary_Sunrise        -0.1088      0.039     -2.815      0.005      -0.185      -0.033
studio_primary_Toei Animation -0.2325      0.034     -6.853      0.000      -0.299      -0.166
genre_Adventure               -0.0783      0.032     -2.467      0.014      -0.140      -0.016
genre_Comedy                  -0.2271      0.052     -4.374      0.000      -0.329      -0.125
genre_Drama                    0.2881      0.040      7.192      0.000       0.210       0.367
==============================================================================
Omnibus:                      143.175   Durbin-Watson:                   1.946
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               68.833
Skew:                           0.052   Prob(JB):                     1.13e-15
Kurtosis:                       2.407   Cond. No.                     2.81e+04
==============================================================================
```

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Limitations of statistical inferences using regression coefficients?

- **Correlated Features:** When predictors are correlated, the individual effect of each one becomes unclear. A change in one feature might be reflected in another, making their coefficient estimates unreliable or misleading.
- **Assumption of Linearity:** Linear regression assumes a straight-line relationship between inputs and the target variable. If the actual relationship is curved or more complex, the coefficients can give an incorrect picture of how features influence the outcome.
- **Lack of Interaction Modeling:** The model considers each feature in isolation, without accounting for the possibility that two or more variables may jointly influence the target in a way that's not captured by looking at them individually.
- **General Rather than Specific Insights:** Coefficients provide an overall view of feature influence across the entire dataset, but they don't help explain individual predictions - why the model made a certain prediction for a specific data point.

These limitations highlight the need for more advanced interpretability tools, which offer both global and instance-level explanations and take into account feature interactions and non-linearity.

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SHAP (SHapley Additive exPlanations) Values

- **SHAP values** help us understand how each feature contributes to a model's prediction for a specific data point. They provide a fair way to divide credit (or blame) among the features for the final prediction.
- **SHAP compares the model's prediction with and without each feature, across all possible combinations of features, to see how much each feature truly influences the result.** It provides a way of testing the model with different groups of features to see the impact of each one.
- Unlike coefficient interpretations for linear regression, **SHAP values work for any model** (not just linear ones), handle feature interactions, and explain individual predictions. They don't get confused by feature correlation or non-linear relationships.
- SHAP also offers different visualizations to gain a better understanding of the contributions of different features to the predictions made by the model.
    - **Waterfall plot:** Shows how each feature pushes a single prediction higher or lower - great for understanding one specific case (local explanation).
    - **Beeswarm plot:** Summarizes how all features influence predictions across the entire dataset - useful for big-picture insights (global explanation).

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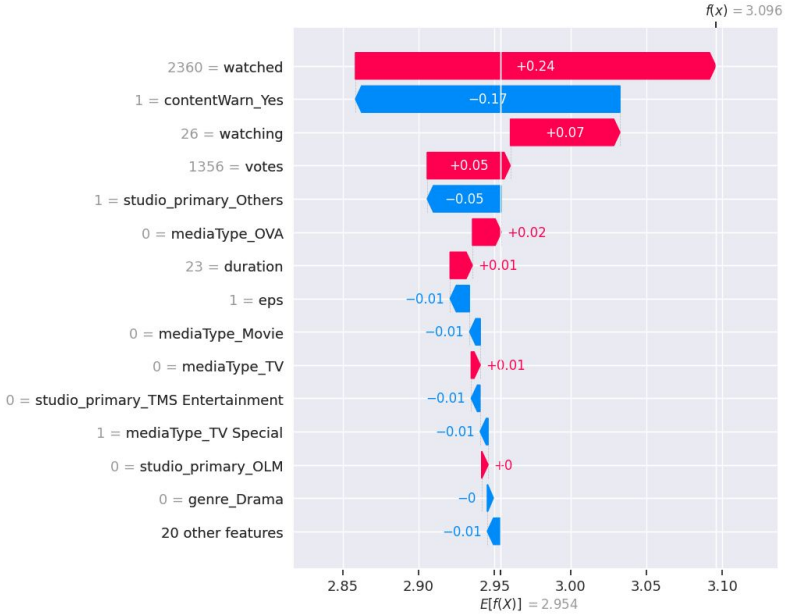
Waterfall plot (Local Explanation)

- A **waterfall plot** shows how each feature pushes a model's prediction up or down from the baseline, making it easy to understand individual predictions.
- **How to read this plot:**
  - Start at the base value, **E[f(X)] = 2.954**
  - Move upward by adding (blue) or subtracting (red) each feature's SHAP value.
  - The final value, **f(x) = 3.096**, at the top is the model's predicted rating for the instance.
- watched = 2360 (blue): Strongly increases the predicted rating by +0.24: more views → higher rating.
- contentWarn_Yes = 1 (red): Significantly decreases the rating by -0.17: presence of content warning lowers appeal.
- Similarly, we can understand how other variables impact the predictions as well.



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Beeswarm plot (Global Explanation)

- A **beeswarm plot** shows how much each feature contributes to the model's predictions across all data points, helping identify the most impactful features and the direction of their influence.
- **How to read this plot:**
  - Each dot represents one observation from the data
  - X-axis: SHAP value - how much the feature pushes prediction higher (right) or lower (left).
  - Y-axis: Features - helps analyze the impact of the feature on a global level.
  - Color: Feature value (red = high, blue = low).
- watched (blue, negative): Lower values strongly reduce prediction, and vice-versa.
- duration (red, positive): Longer watch durations are associated with higher ratings.
- Similarly, we can understand how other variables impact the predictions as well.

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Power Ahead!

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